

Nandhakumar V

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AI & DATA SCIENCE PROFESSIONAL

AI & Data Science graduate with verified hands-on exposure spanning data science, data analytics, machine learning, deep learning, NLP, data processing, SQL/database work and Python backend development. Suitable for entry-level Data Scientist, Data Analyst, Data Engineer, Python/Full Stack Developer and Junior AI/ML roles, with academic team work clearly separated from individual technical experience.

Education

Dhirajlal Gandhi College Of Technology

2021 – 2025

B.Tech – Artificial Intelligence and Data Science | CGPA: 7.8

Work History

Data Science Consultant, Intern | Rubixe – AI Solutions

Bengaluru (Onsite) | Jun 2025 – Jan 2026

- Developed and evaluated machine-learning workflows using Python and Scikit-learn for classification and regression tasks.
- Applied EDA, statistical analysis, preprocessing and feature engineering on structured datasets for data-driven analysis.
- Contributed to the IT Incident Management client project using approximately 46K enterprise records for incident prioritisation.
- Completed the Certified Data Scientist programme through the DataMites learning pathway alongside project and practical work.

Full Stack Developer | Aggregate Intelligence Inc

Coimbatore (Onsite) | Aug 2024 – Jan 2025

- Worked onsite on full-stack application development, contributing to frontend/backend integration and application workflows.
- Developed Python backend components and REST APIs for application and data-integration workflows.
- Worked with MySQL/database operations and SQL query optimisation, with approximately 20% lower query latency.
- Used Git/GitHub and Docker in application development, testing and integration workflows.

Data Science Course & Projects | EdiGlobe Pvt Ltd

Bengaluru (Virtual) | Aug 2024 – Oct 2024

- Completed structured self-paced Data Science learning and project work using Python and data-analysis libraries.
- Performed data cleaning, EDA, statistical analysis and visualization using Pandas, NumPy, Matplotlib and Seaborn.
- Applied preprocessing and feature engineering to prepare datasets for predictive modelling and analysis.
- Completed House Price Prediction and IMDb Movie Data Analysis projects.

Skills

- **Programming & Data:** Python, SQL, MySQL, Pandas, NumPy, Matplotlib, Seaborn
- **Data Science & Analytics:** Data Collection, Data Cleaning, EDA, Statistical Analysis, Descriptive Statistics, Correlation Analysis, Trend Analysis, Pattern Recognition, Data Visualization, Data Profiling, Data Validation, Feature Engineering, Data Preprocessing, Model Evaluation
- **Machine Learning:** Classification, Regression, Logistic Regression, Linear Regression, KNN, SVM, Decision Trees, Random Forest, XGBoost, LightGBM, CatBoost, K-Means, PCA
- **Deep Learning & NLP:** TensorFlow, Keras, CNN, LSTM, ANN, Transformers, NLTK, NLP, Text Preprocessing, Sentiment Analysis, Image Classification, Computer Vision
- **SQL & Data Engineering:** SQL Query Writing, Query Optimisation, SQL Joins, Aggregations, Filtering, Data Extraction, Data Transformation, Data Validation, Data Pipelines, SQLAlchemy, Numba JIT, Large-Dataset Processing
- **Backend & Software:** Python Backend Development, Flask, REST APIs, API Integration, Frontend/Backend Integration, Git, GitHub, Docker, Application Testing and Integration
- **Training / Basics:** Excel, Power BI, Tableau; FastAPI (Basics), Linux, AWS EC2 (Basics), PyTorch (Basics), Prompt Engineering, LLM concepts

Certifications:

- **Certified Data Scientist — NASSCOM / FutureSkills Prime** | Gold Category | 91% Assessment Score | Apr 2026
- **Data Science Foundation — IABAC** | Effective May 2026
- **Certified Data Scientist (CDS-DS2050) — DataMites** | 700+ hours | IABAC / NASSCOM accredited
- **Data Science Foundations – Level 2 (V2) — IBM / Credly** | Sep 2025
- **Data Science 101 — IBM** | Sep 2025
- **Data Science Tools — IBM** | Sep 2025
- **Data Science Methodology — IBM** | Sep 2025
- **Python (Basic) — HackerRank** | Sep 2025

Projects

IT Incident Management using Machine Learning

- Analysed approximately 46K enterprise ITSM records from a live MySQL database for incident prioritisation and operational analytics.
- Goal: improve ticket triage and prioritisation and reduce manual effort in IT incident management.
- Applied SQLAlchemy, data cleaning, ITIL Impact × Urgency priority logic, feature engineering, Random Forest, Prophet and Gradient Boosting with SMOTE across separate workflows.
- Verified results included High-Priority Random Forest F1-score of approximately 0.98, Auto-Tagging around 83% and RFC-Failure around 93%.

Skills Applied: Python | SQL | MySQL | SQLAlchemy | EDA | Feature Engineering | Scikit-learn | Random Forest

AI-Powered Sentiment Analysis

- Built a customer-review sentiment analysis workflow using the Amazon Datafiniti dataset containing 5,000 rows and 24 columns.
- Goal: classify customer sentiment automatically and support scalable analysis of review feedback.
- Applied text preprocessing, NLTK, TensorFlow/Keras and Transformers, with comparative modelling including LSTM and Random Forest.
- Verified results: Random Forest 94.8% accuracy and LSTM 93.7%.

Skills Applied: Python | NLP | NLTK | TensorFlow | Keras | LSTM | Transformers | Sentiment Analysis

Home Loan Default Risk Management

- Developed a machine-learning workflow using Home Credit data; application_train contains 307,511 rows, with seven CSV sources used in the broader analysis.
- Goal: identify customers at risk of loan default and support data-driven credit-risk assessment.
- Applied multi-table integration, preprocessing, memory-aware optimisation including Numba JIT, feature engineering and model comparison; SMOTE was used only for the RFC sub-model.
- Verified result: XGBoost was the strongest reported model at approximately 92% accuracy; XGB AUC was 0.760.

Skills Applied: Python | Pandas | Scikit-learn | XGBoost | Feature Engineering | Data Preprocessing

Skin Disorder Detection

- Built a multiclass classification workflow using a dermatology dataset with 366 rows, 35 features and six disorder classes.
- Goal: classify skin-disorder categories from structured clinical attributes using machine-learning methods.
- Applied preprocessing and compared Logistic Regression, KNN and Random Forest using classification metrics.
- Verified test accuracy: Random Forest 100%, Logistic Regression 98.65%, KNN 85.14% and SVM 75.68%.

Skills Applied: Python | Scikit-learn | Classification | Logistic Regression | KNN | SVM | Random Forest

Rice Leaf Disease Detection

- Built a CNN-based image-classification workflow using 119 rice-leaf images across Bacterial Blight, Brown Spot and Leaf Smut.
- Goal: demonstrate automated rice-leaf disease classification from images.
- Used 128×128 image processing, an 80/20 split, ImageDataGenerator augmentation and TensorFlow/Keras CNN model comparison.
- Verified best model accuracy was 71%.

Skills Applied: Python | TensorFlow | Keras | CNN | Image Classification | Data Augmentation

Earthquake Damage Prediction

- Developed a multiclass classification workflow for the DrivenData earthquake-damage dataset using structural and geographical features.
- Goal: predict building damage categories for data-driven post-earthquake risk assessment.
- Performed data merging, EDA, IQR-based outlier capping and OneHotEncoding; compared Logistic Regression, Decision Tree, Random Forest, XGBoost, LightGBM and CatBoost.
- No unsupported final accuracy is stated because a reliable final figure was not captured in the verified notebook.

Skills Applied: Python | Pandas | Scikit-learn | XGBoost | LightGBM | CatBoost | EDA

House Price Prediction

- Built a regression model using the Boston Housing dataset containing 506 observations and 13 features.
- Goal: estimate housing prices from structured property features.
- Applied missing-value handling, StandardScaler preprocessing and Linear Regression with regression evaluation.
- Verified performance: $R^2 = 0.6589$ and $MAE = 3.1499$.

Skills Applied: Python | Pandas | Scikit-learn | Regression | Linear Regression | StandardScaler

IMDb Movie Data Analysis

- Performed exploratory analysis on an IMDb movie dataset containing 1,000 movies and 12 columns.
- Goal: identify movie trends, relationships and data-quality patterns across ratings, revenue, votes, runtime and related fields.
- Performed 22 EDA tasks covering cleaning, missing-value checks, statistical summaries, correlations, distributions and visual analysis.
- This is an EDA project only; no sentiment-classification model is claimed.

Skills Applied: Python | Pandas | NumPy | EDA | Statistical Analysis | Data Visualization